

We will cover only the following three topics in the Week 11 lecture.

- 1 Surface orientation.
- 2 Surface integrals of vector fields.
- 3 Flux.

We now turn to surface integrals of vector fields.

To evaluate such surface integrals, we need to fix a **surface orientation**.

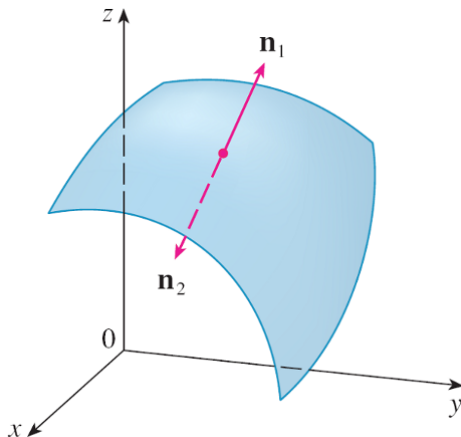
Definition (Orientation)

A smooth parametric surface S is said to be **orientable** (or just **two-sided**) if and only if there exists an $\mathbb{R}^3$ -valued vector function $\mathbf{n}$ defined throughout the surface interior of S that satisfies the following two properties:

- For every point p within the surface interior of S , the vector $\mathbf{n}(p)$ is a unit normal vector to S at p .
- $\mathbf{n}$ is continuous.

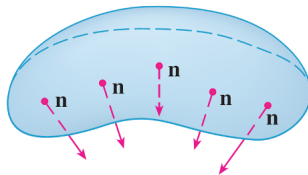
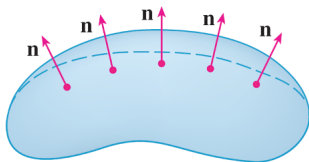
Surface Integrals — Surface Orientation

At each surface interior point of a smooth parametric surface, there exist only two unit normal vectors.



Surface Integrals — Surface Orientation

If a smooth parametric surface S is orientable, then we can assign a unit normal vector to each surface interior point of S so that these vectors vary continuously.



We call such a ~~continuous~~ continuous assignment of unit normal vectors an “**orientation**” of S .

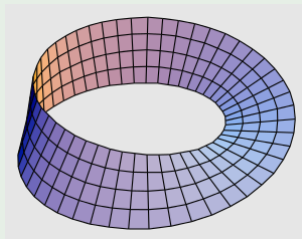
Example (A Non-Orientable Nice and Smooth Parametric Surface)

Define a function $\mathbf{r} : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by

$$\mathbf{r}(u, v) \stackrel{\text{df}}{=} \left(\left(1 + \frac{v}{2} \cos\left(\frac{u}{2}\right) \right), 1 + \frac{v}{2} \cos\left(\frac{u}{2}\right), \frac{v}{2} \sin\left(\frac{u}{2}\right) \right).$$

Let S denote the nice smooth parametric surface parametrized by $(\mathbf{r}, [0, 2\pi] \times [-1, 1])$.

Then S , known as a **Möbius strip**, is non-orientable (it has only one side)!



*✗ we won't be dealing
with non-orientable
surfaces*

Definition (Oriented Smooth Parametric Surfaces)

A **oriented** smooth parametric surface is an orientable smooth parametric surface with a choice of orientation.

★ Moving forward, we will deal only with orientable smooth parametric surfaces.

The following are well-known orientable smooth parametric surfaces:

- Planes.
- Spheres.
- Ellipsoids.
- Cylinders.
- Paraboloids.
- Hyperboloids.

Surface Integrals — Surface Orientation

If S is an orientable smooth parametric surface, then define an orientation $\mathbf{n}$ as follows:

- 1 Find a smooth surface parametrization $(\mathbf{r}, D)$ of S .
- 2 Given $(u, v) \in D^\circ$, let

$$\mathbf{n}(\mathbf{r}(u, v)) \stackrel{\text{df}}{=} \frac{1}{\|(\mathbf{r}_u \times \mathbf{r}_v)(u, v)\|} \cdot (\mathbf{r}_u \times \mathbf{r}_v)(u, v). \quad \left. \vphantom{\frac{1}{\|(\mathbf{r}_u \times \mathbf{r}_v)(u, v)\|}} \right\} \begin{array}{l} \text{unit normal} \\ \text{vector} \end{array}$$

Caution: $\mathbf{n}$ is defined on a subset of S , not on D° !!! $\Rightarrow$ on surface, not domain!

- 3 Given $(u, v) \in D^\circ$, we can also let

$$\mathbf{n}(\mathbf{r}(u, v)) \stackrel{\text{df}}{=} -\frac{1}{\|(\mathbf{r}_u \times \mathbf{r}_v)(u, v)\|} \cdot (\mathbf{r}_u \times \mathbf{r}_v)(u, v).$$

- 4 Which surface orientation to choose will be specified by the problem.

Surface Integrals — Surface Orientation (Upward and Downward)

When the smooth parametric surface in question is a graph surface, then we can select a very natural orientation for it.

Let D be a subset of $\mathbb{R}^2$ with D° non-empty and connected.

Let g be a continuous scalar function defined on D with the following two properties:

- ∇g is defined and continuous throughout D° .
- ∇g is bounded on D° .

Then define an orientation $\mathbf{n}$ on $\text{Graph}(g)$ so that, given any $(x, y) \in D^\circ$,

$$\mathbf{n}(x, y, g(x, y)) \stackrel{\text{df}}{=} \frac{1}{\sqrt{1 + [g_x(x, y)]^2 + [g_y(x, y)]^2}} \cdot (-g_x(x, y), -g_y(x, y), \boxed{1}).$$

We call this orientation of $\text{Graph}(g)$ the “**upward orientation**”.

The opposite orientation, $\boxed{-\mathbf{n}}$, is called the “**downward orientation**”.

$$-\vec{n}(x, y, g(x, y)) = \frac{1}{\sqrt{1 + [g_x(x, y)]^2 + [g_y(x, y)]^2}} (g_x(x, y), g_y(x, y), -1)$$

Surface Integrals — Surface Orientation (Positive and Negative)

Let E be a connected open subset of $\mathbb{R}^3$.

Suppose that the boundary of E is a piecewise-smooth surface S .

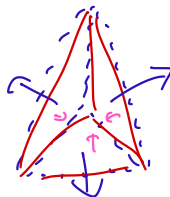
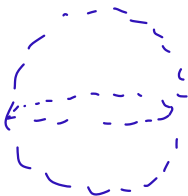
Then each of the smooth parametric surfaces that form S can be given the orientation for which all unit normal vectors point outward.

This equips S with an overall orientation, which we call the “outward! **positive orientation of S** ”.

The opposite orientation is the one for which all unit normal vectors point inward, and is called the “**negative orientation of S** ”.

for curves, positive means
anti clockwise

inward



normal vectors!

Definition (Surface Integral of a Vector Field)

- Let S be an oriented, nice, and smooth parametric surface, with orientation $\mathbf{n}$.
- Let $\mathbf{F}$ be a continuous space vector field that is bounded on $S \subseteq \text{Dom}(\mathbf{F})$.

Define the **surface integral of $\mathbf{F}$ over S** , denoted by $\iint_S \mathbf{F} \bullet d\mathbf{S}$, as follows:

$$\boxed{\iint_S \mathbf{F} \bullet d\mathbf{S}} \stackrel{\text{df}}{=} \iint_S \boxed{\mathbf{F} \bullet \mathbf{n}} dS$$

Handwritten annotations: "Expression" above the first box, "scalar field" above the second box, "bold S" below the first box, "Tested" to the right of the second box, and "italic S" below the second box.

In physics, the quantity $\iint_S \mathbf{F} \bullet d\mathbf{S}$ is called the "**flux of $\mathbf{F}$ across S** ". It measures how much of a force field is flowing perpendicularly across S in a given direction.

different!

Surface Integrals — Vector Fields

The formula in the preceding definition can be made more explicit.

Let $(\mathbf{r}, D)$ be an orientation-respecting nice surface parametrization of S .

This means that, given $(u, v) \in D^\circ$,

$$\mathbf{n}(\mathbf{r}(u, v)) = \frac{1}{\|\mathbf{r}_u(u, v) \times \mathbf{r}_v(u, v)\|} \cdot [\mathbf{r}_u(u, v) \times \mathbf{r}_v(u, v)]. \text{ from earlier on}$$

Then What to find

$$\begin{aligned} \boxed{\iint_S \mathbf{F} \bullet d\mathbf{S}} &= \iint_S \mathbf{F} \bullet \mathbf{n} \, dS \\ &= \iint_{D^\circ} [\mathbf{F}(\mathbf{r}(u, v)) \bullet \mathbf{n}(\mathbf{r}(u, v))] \cdot \|\mathbf{r}_u(u, v) \times \mathbf{r}_v(u, v)\| \, dA(u, v) \\ &= \iint_{D^\circ} \mathbf{F}(\mathbf{r}(u, v)) \bullet [\mathbf{r}_u(u, v) \times \mathbf{r}_v(u, v)] \, dA(u, v) \\ &= \iint_{D^\circ} [(\mathbf{F} \circ \mathbf{r}) \bullet (\mathbf{r}_u \times \mathbf{r}_v)](u, v) \, dA(u, v). \end{aligned}$$

} surface integral of vector field

↖ use this formula

Caution: $(\mathbf{r}, D)$ must be orientation-respecting!!!

Example

- Let S denote the surface boundary of the closed region in $\mathbb{R}^3$ that is
 - bounded above by the paraboloid given by the equation $z = 1 - x^2 - y^2$;
 - bounded below by the xy -plane.
- Let S be oriented positively.
- Define a continuous space vector field on $\mathbb{R}^3$ by $\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (y, x, z)$.

Problem: Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$.

Solution

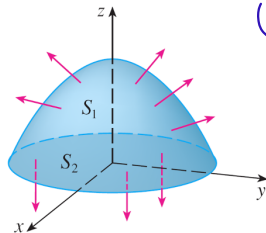
$$(S_1) \{ (x, y, 1 - x^2 - y^2) \in \mathbb{R}^3 \mid x^2 + y^2 \leq 1 \}$$

$$(S_2) \{ (x, y, 0) \in \mathbb{R}^3 \mid x^2 + y^2 \leq 1 \}$$

Define $\vec{r}_1, \vec{r}_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by

$$\vec{r}_1(r, \theta) = \langle r \cos \theta, r \sin \theta, 1 - r^2 \rangle$$

$$\vec{r}_2(r, \theta) = \langle r \cos \theta, r \sin \theta, 0 \rangle$$



Surface Integrals — Example 3 (cont'd)

For all $r, \theta \in \mathbb{R}$, we have

$$\left. \begin{aligned} \cdot \frac{\partial \vec{r}_1}{\partial r}(r, \theta) &= \langle \cos \theta, \sin \theta, -2r \rangle \\ \cdot \frac{\partial \vec{r}_1}{\partial \theta}(r, \theta) &= \langle -r \sin \theta, r \cos \theta, 0 \rangle \end{aligned} \right\} \left(\frac{\partial \vec{r}_1}{\partial r} \times \frac{\partial \vec{r}_1}{\partial \theta} \right)(r, \theta) = \langle 2r^2 \cos \theta, 2r^2 \sin \theta, r \rangle$$

→ correct orientation

$$\left. \begin{aligned} \cdot \frac{\partial \vec{r}_2}{\partial r}(r, \theta) &= \langle \cos \theta, \sin \theta, 0 \rangle \\ \cdot \frac{\partial \vec{r}_2}{\partial \theta}(r, \theta) &= \langle -r \sin \theta, r \cos \theta, 0 \rangle \end{aligned} \right\} \left(\frac{\partial \vec{r}_2}{\partial r} \times \frac{\partial \vec{r}_2}{\partial \theta} \right)(r, \theta) = \langle 0, 0, r \rangle$$

Wrong orientation $\Rightarrow S_2$ point down

$(\vec{r}_1, [0, 1] \times [0, 2\pi])_i$ is a parameterization of S_1 .

$(\vec{r}_2, [0, 1] \times [0, 2\pi])_i$ is a parameterization of S_2 .

but r is positive

→ $0 \leq r \leq 1$

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \iint_{[0, 1] \times [0, 2\pi]} \vec{F}(\vec{r}_1(r, \theta)) \cdot \left(\frac{\partial \vec{r}_1}{\partial r} \times \frac{\partial \vec{r}_1}{\partial \theta} \right)(r, \theta) dA(r, \theta) \\ &= \iint_{[0, 1] \times [0, 2\pi]} \langle r \sin \theta, r \cos \theta, 1 - r^2 \rangle \cdot \langle 2r^2 \cos \theta, 2r^2 \sin \theta, r \rangle dA(r, \theta) \\ &= \iint_{[0, 1] \times [0, 2\pi]} [2r^3 \sin(2\theta) - r^3 + r] dA(r, \theta) \\ &= \int_0^{2\pi} \left[\int_0^1 [2r^3 \sin(2\theta) - r^3 + r] dr \right] d\theta \\ &= \int_0^{2\pi} \left[\frac{r^4}{2} \sin(2\theta) - \frac{r^4}{4} + \frac{r^2}{2} \right]_{r=0}^{r=1} d\theta \\ &= \int_0^{2\pi} \left[\frac{1}{2} \sin(2\theta) + \frac{1}{4} \right] d\theta \end{aligned}$$

Surface Integrals — Example 3 (cont'd)

$$= \left[-\frac{1}{4} \cos(2\theta) + \frac{1}{4} \theta \right]_0^{2\pi}$$

$$= \frac{\pi}{2}$$

$$\iint_{S_2} \vec{F} \cdot d\vec{S} = - \iint_{-S_2} \vec{F} \cdot d\vec{S}$$

$$= - \iint_{[0,1] \times [0,2\pi]} \vec{F}(\vec{r}_2(r,\theta)) \cdot \left(\frac{\partial \vec{r}_2}{\partial r} \times \frac{\partial \vec{r}_2}{\partial \theta} \right)(r,\theta) \, dA(r,\theta)$$

$$= - \iint_{[0,1] \times [0,2\pi]} \langle r \sin \theta, r \cos \theta, 0 \rangle \cdot \langle 0, 0, r \rangle \, dA(r,\theta)$$

$$= - \iint_{[0,1] \times [0,2\pi]} 0 \, dA(r,\theta)$$

$$= -0$$

$$= 0$$

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_{S_1} \vec{F} \cdot d\vec{S} + \iint_{S_2} \vec{F} \cdot d\vec{S} = \frac{\pi}{2} + 0 = \frac{\pi}{2}$$

Example

- Let S denote the top hemisphere with center $(0, 0, 0)$ and radius 6.
- Let S be oriented upwards.
- Define a continuous space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by $\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (0, 0, z)$.

Problem: Find the flux of $\mathbf{F}$ across S .

Solution

Example

- Let S denote the surface boundary of the closed region in $\mathbb{R}^3$ that is
 - bounded above by the top hemisphere given by the equation $z = \sqrt{8 - x^2 - y^2}$;
 - bounded below by the closed cone given by the equation $z = \sqrt{x^2 + y^2}$.
- Let S be oriented positively.
- Define a continuous space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by $\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (y, 0, 2)$.

Problem: Find the flux of $\mathbf{F}$ across S .

Solution

